COMPUTER-AIDED MACHINE DRAWING

23A03305 – Exercises with Solutions

II Year B.Tech ME – I Semester, JNTUA R23 (L 0, T 0, P 3, C 1.5)

Course objectives: conventional representation of materials and machine components; 2D and 3D modelling software; thread profiles, riveted, welded and key joints; solid modelling of machine parts and their sections; 2D and 3D assembly drawings; limits, fits and tolerances of mating components.

Course outcomes

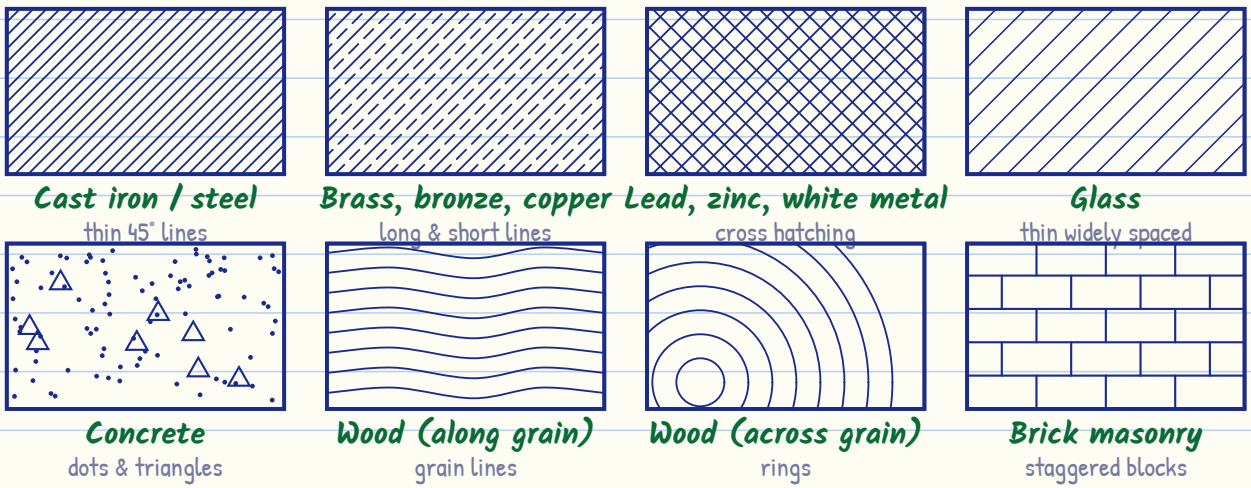
CO	Statement	Blooms
CO1	Demonstrate the conventional representations of materials and machine components	L3
CO2	Model riveted, welded and key joints using CAD system	L6
CO3	Create solid models and sectional views of machine components	L6
CO4	Generate solid models of machine parts and assemble them	L5
CO5	Translate 3D assemblies into 2D drawings	L6

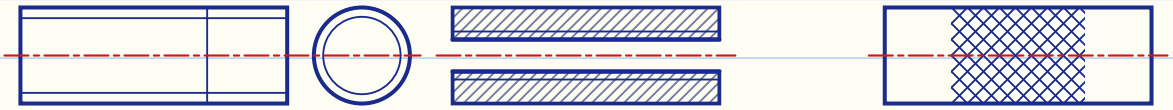
Plan of the record: Part A – 2D package (AutoCAD / any 2D software) exercises 1 – 14. Part B – 3D package (SolidWorks / CATIA / Creo / Inventor) sectional views and assembly drawings. Part C – production drawing with limits, fits and tolerances. In every exercise: d = nominal (shank) diameter, all other sizes in terms of d, as in the textbooks.

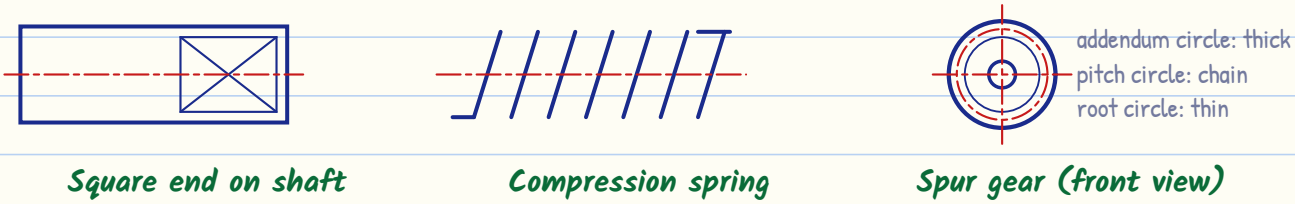
PART A – 2D PACKAGE

1. Conventional Representation

Conventions (IS 696 / SP 46) save the drawing time: a section is hatched with thin continuous lines at 45° (spacing 2 – 3 mm), and the material is shown by the type of hatching. Components like threads, springs, knurling, gears are drawn in a simplified way.





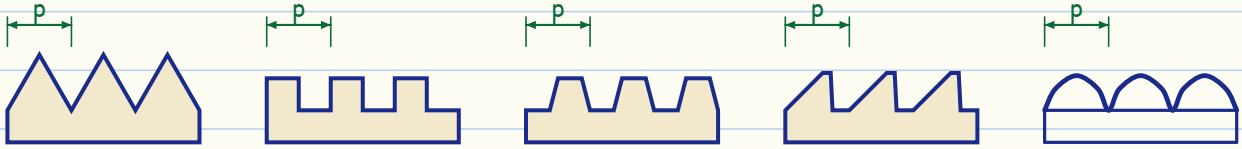


Answer - points to remember

- Hatching: thin lines at 45° to the axis or the outline, equally spaced; in the assembly adjacent parts hatched in opposite directions (or at different spacing); never hatch bolts, nuts, shafts, keys, rivets, webs and ribs when cut longitudinally.
- External thread: major diameter by a thick line, minor (root) diameter by a thin line (about 0.8 d); in the end view the root is a thin 3/4 circle. Internal thread (section): the thick line for the minor diameter, thin for the major.
- Knurling: a pattern of crossed (diamond) or straight lines drawn near the ends of the area. Square end: shown by thin diagonals. Spring: a continuous line replaces the coils, with a chain line through the centre of the coil, only a few coils shown.
- Spur gear: addendum circle thick, pitch circle chain thin, root circle thin continuous or omitted; in the section the teeth are not hatched.
- Line types: continuous thick (visible outline), continuous thin (dimension, hatch, projection), dashed (hidden), chain thin (centre lines), chain thick at ends (cutting plane).

2. Thread Profiles

Thread = a helical ridge of the uniform section on a cylindrical surface. Pitch p = distance between the corresponding points of adjacent threads; lead = pitch x number of starts.



Sharp V
60°, ISO metric
fasteners

Square
90° flanks
vice, press

Acme
29° included
lead screw

Buttress
45° + 3°
one-way load

Knuckle
rounded
hydrant, bulb

Profile	Angle	Depth / proportions	Use
Sharp V	60°	depth = 0.866 p	theory only (sharp crest and root)
ISO metric (basic)	60°	depth = 0.6134 p, crest and root flats p/8 and p/4	fasteners: bolts, nuts, studs
B.S.W. (Whitworth)	55°	depth = 0.64 p, rounded crest and root, r = 0.137 p	pipes, general
Square	90° flanks	depth = p/2, width = p/2	power screws, vices, presses
Acme	29° included	depth = p/2 + 0.25 mm, crest width 0.3707 p	lead screw of lathe, valve spindle
Buttress	45° and 7°	depth = 0.75 p	one way thrust: screw jack, vice
Knuckle	rounded	depth = 0.5 p, radius = p/4	bulbs, hydrant, railway coupling

Steps (2D software)

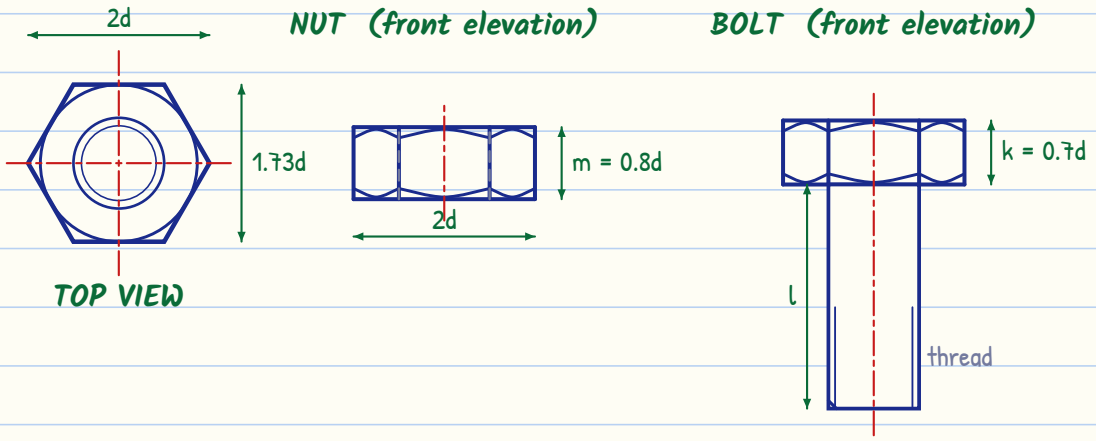
- Draw the centre line. With OFFSET make the lines at a distance p/2 on both sides (the 'pitch line'), draw the construction line at p intervals with ARRAY.
- Draw the profile for one pitch with LINE and polar tracking (30° for sharp V, 14.5° for Acme), MIRROR and ARRAY for 4 - 5 threads, TRIM the extra lines. Dimension: pitch p, depth, angle, major and minor diameters.

3. Detachable Joints - Bolts and Nuts

Detachable (temporary) joints can be taken apart without damage to the parts: bolts, studs, screws, keys, cotters, couplings. The hexagonal bolt and nut is the most common.

Proportions (d = nominal diameter of bolt)

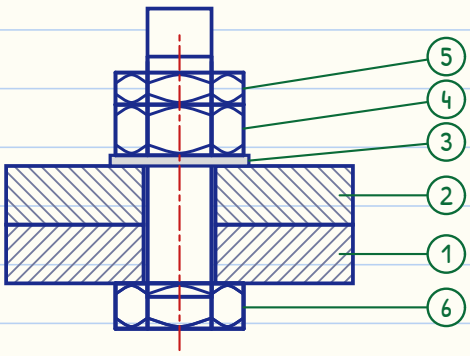
Part	Proportion
Hexagonal nut: thickness m	0.8 d (square nut: d)
Width across the corners (hex nut / head)	2 d (across flats = 1.75 d approx.)
Bolt head thickness k	0.7 d
Square head (bolt / nut)	side = 1.5 d (across the corner 2.1 d), 0.7 d thick for the bolt head, d for the nut
Washer	outer dia = 2.2 d, thickness 0.15 d, hole d + 1.5 mm
Lock nut (thin jam nut)	thickness 0.5 d, fitted over the main nut
Chamfer	30° on the corners; the chamfer arcs are seen on the three faces as the arcs on the front view
Bolt length of thread	2 d + 6 mm for l up to 125 mm; the bolt end is chamfered



Steps for the hexagonal nut (view from the front)

- Draw the top view first: circle of dia 2d (circumscribing the hexagon), and the chamfer circle of dia = across flats (1.75d). Draw the hexagon in it (corners on the horizontal line).
- Project up the front view: the nut height m = 0.8 d; draw the three faces (the corner distance projected). The chamfer 30° produces the arcs with radii R = across flats approx, located on the faces. Draw the circular hole d (thread: minor circle 0.8d, thin 3/4 arc).
- When the across flats is facing the viewer, only the two faces are seen (the 'elevation from the flats' view), width 1.75d.

Bolted joint with washer and lock nut



1 Plate (lower)
2 Plate (upper)
3 Washer
4 Nut
5 Lock nut
6 Bolt

k = 0.7d, nut m = 0.8d
lock nut = 0.5d
washer: OD 2.2d, t = 0.15d
hole = d + 1.5 mm
across corners 2d

- Two plates are clamped by a bolt, the head on one side, washer + nut + lock nut on the other. Hole = d + 1.5 mm. The bolt end projects about 0.25 d to 1 d beyond the lock nut.
- In the assembly the bolt, nut and washer are not hatched in the section; plates hatched in opposite directions.
- Lock nut: thin nut (0.5 d) is screwed on the standard nut (0.8 d) and tightened against it - the two nuts jam against each other and prevent the loosening from the vibration (also split pin, castle nut, spring washer, Grover washer, Sasa nut).

Example

Draw the two views of a M20 hexagonal bolt with nut and washer for the plates 20 mm and 25 mm thick.

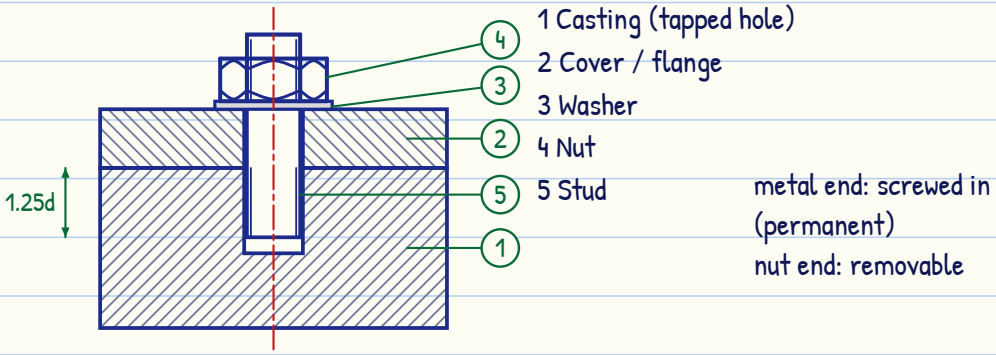
- d = 20: head k = 14, nut m = 16, across corners 40 mm, across flats 35 mm, washer OD 44, t = 3, hole 21.5 mm.
- Grip length = 20 + 25 = 45 mm; bolt length l = 45 + 3 (washer) + 16 (nut) + 0.25d ~ 5 -> l = 69 ~ 70 mm (standard length 70). Thread length 2d + 6 = 46 mm.

4. Stud Joint and Screw (Set / Cap Screw) Joint

Stud joint

A stud is a rod threaded on both ends: one end is screwed permanently into the tapped hole of one of the parts (e.g. the cylinder head, casing) and the other end carries the nut and washer

to fasten the cover.



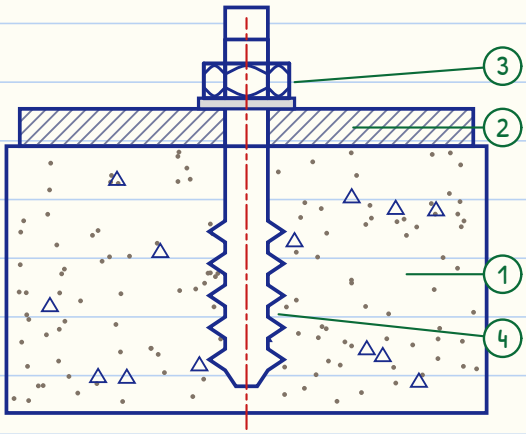
- Proportions: length of the threaded end screwed in = 1.25 d to 2 d (1.5 d in the cast iron, d in the steel); depth of the tapped hole 2d; the nut end projects 0.25 d to 1 d beyond the nut.
- The hole in the cover = d + 1.5 mm (clearance), and the tapped hole is shown with the minor diameter thick and the major diameter thin. The drill depth beyond the thread = 0.5d (the cone of 118° at the bottom).
- Stud is not removed during dismantling, only the nut. This saves the threads of the casting.

Screw joint (cap screw / set screw)

- A cap screw passes through the clearance hole of the top part and is screwed into the tapped hole of the bottom. Hexagonal or socket head cap screw head: k = 0.7 d (cheese head: dia 1.5d, height 0.7d; countersunk: dia 2d, 90° cone; pan head dia 1.75d). A set screw (grub screw) locks collars on shafts - dog point, cup point, flat point.
- The slot of a screw is drawn at 45° in the top view, width 0.2 d - 0.25 d, depth 0.25 d.

5. Foundation Bolts

Foundation bolts hold the machine base plates to the concrete foundation: rag (barbed) bolt, lewis bolt, eye bolt, cotter bolt, curved bolt, square-headed bolt with plate (washer).



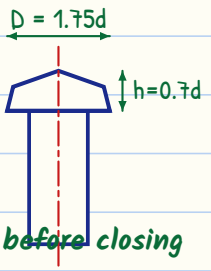
1 Concrete foundation
2 Machine base plate
3 Washer and nut
4 Rag bolt (barbed end)

Rag bolt: barbs or rag cuts grip the concrete / lead fill. Bolt is placed before the concrete is poured.

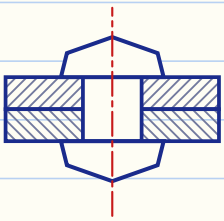
- Rag bolt: the barbs (rough, cut ends) are held by cement-concrete poured around it; the bolt is placed in the foundation before concreting; head is a nut and washer.
- Lewis bolt: a tapered bolt with the key, put into a dovetail pocket of the foundation, and grouted with lead or cement.
- Eye bolt: the bolt with a cross-pin (rod) in the eye, held in a pocket of the concrete.
  Cotter bolt: a cotter or key at the bottom against the plate gives the grip.
- Proportions: length of the bolt in the concrete = 20d to 25d (min.); the bolt is 'd' diameter, the washer / plate under the nut OD = 2.2d. The base plate hole = d + 1.5mm.
- The foundation concrete shown with the dots and triangles (conventional), the machine base in cast iron hatch.

6. Riveted Joints

A rivet is a short cylindrical bar with the head on one end, used for permanent joints of plates (boilers, tanks, structures). The shank passes through the holes in the plates, and the second head (closed head or point) is made by hammering the projecting end (about 1.5 d length, for the snap head).



rivet before closing



closed rivet in plates



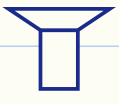
Snap



Pan



Conical



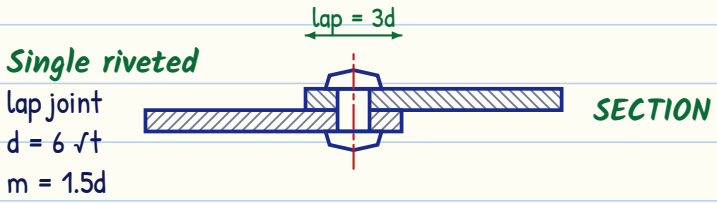
Counter-sunk

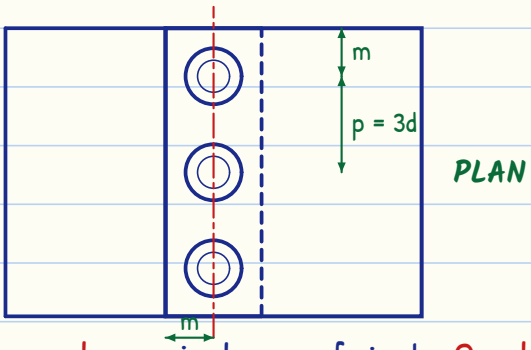
Head type	Proportions
Snap head (round)	dia D = 1.75 d, height 0.7 d (sphere R = 0.9 d)
Pan head	dia D = 1.75 d, height 0.7 d with a cone, the radius at the top 0.4d
Conical head (point)	dia 1.5 d at the base, 0.75d at top, height 0.6d - 0.7 d
Countersunk	90° countersink, dia 1.5 d at top, flush with the plate

Rivet size

- Unwin's formula: d = 6 √t (t = plate thickness in mm); the hole dia = d + 1.5 mm (up to 25 mm rivet, then +2 mm). Pitch (p) = distance between the centres of the adjacent rivets in a row = 3d (min 2d). Margin (m) = distance from the plate edge to the centre of the rivet = 1.5 d. Back pitch pb = distance between the rows (chain 2d - 2.5d, zig-zag 0.33p + 0.67d).

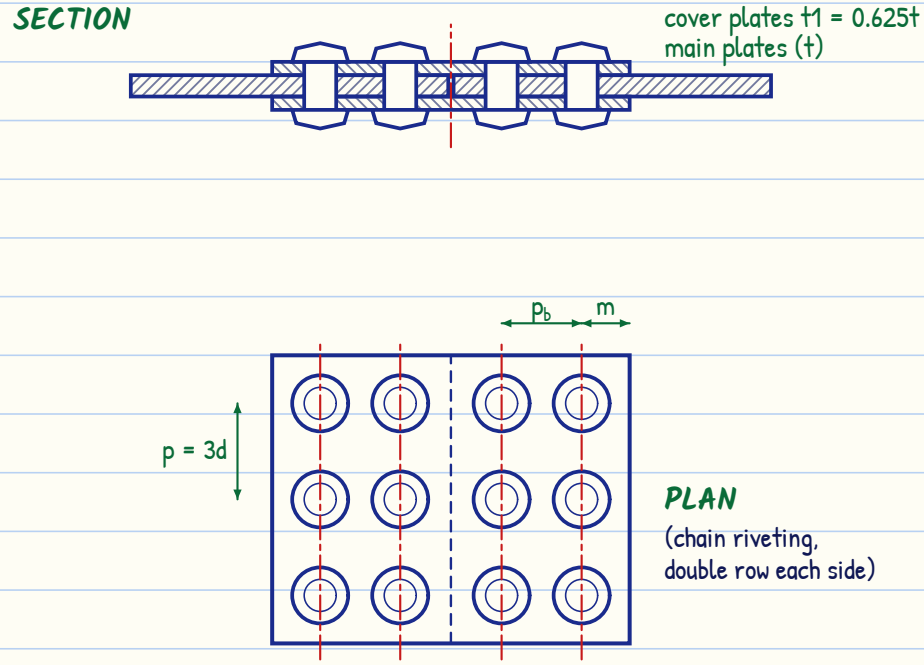
Lap joint (single riveted)





- The plates overlap; a single row of rivets. Overlap = 3d (margin 1.5 d on each side of the rivet row). In double riveted lap joint the overlap = 1.5d + back pitch + 1.5d; chain: pb = 2.5d, zig-zag pb = 0.33p + 0.67d.
- The section (front view) is cut at the rivet centre: the rivets are not hatched (rivet is solid), the plates hatched in the opposite directions. The plan shows the heads (circles dia 1.75d) and the hole centre lines.

Butt joint



- Butt joint with single strap (cover plate on one side, thickness t1 = 1.125 t) and the double strap (both sides, t1 = 0.625 t, i.e. 5/8 of the main plate). The plates meet in the end to end with the small gap, and the cover plates are fixed by the rivets in the rows on either side.
- Single riveted: one row each side; double riveted: two rows each side, chain or zig-zag. The double strap joint has the rivets in double shear - 1.75 times stronger than in the single shear - and no bending of the plate.
- Width of the cover plate: single riveted = 2 x 1.5d (margins) + the gap, about 6d in all; double riveted = 2 margins (1.5d each) + 2 x back pitch + the gap between the plates.

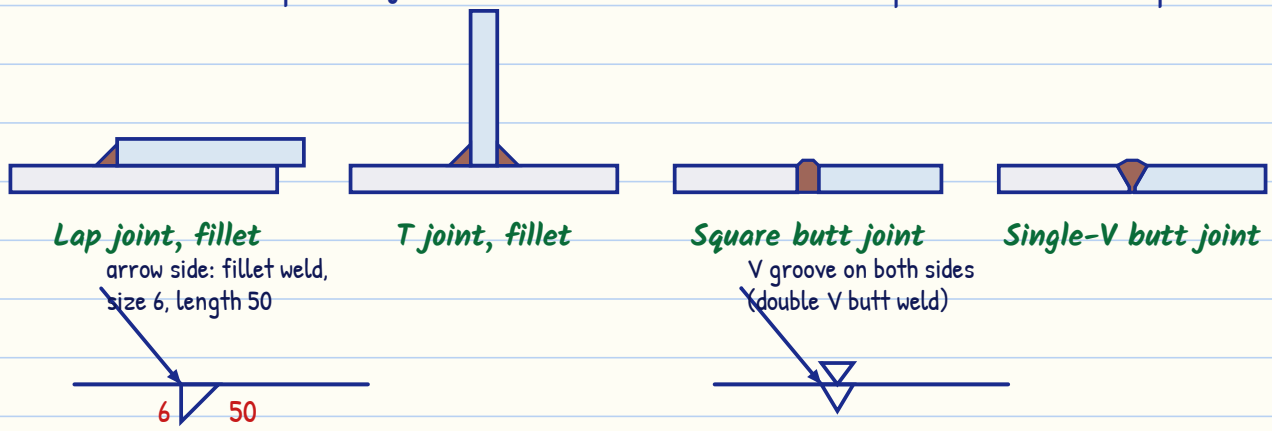
Example

Draw the sectional elevation and plan of a double riveted double strap chain butt joint for the plates 12 mm thick.

- d = 6 √12 = 20.8, so take the rivet d = 21 mm (hole 22.5 mm). p = 3d = 63 (taking 65 mm); pb = 2.5 d = 52; m = 1.5d = 31.5 (32 mm); t1 = 0.625 x 12 = 7.5 mm. Snap head: D = 1.75 d = 37, h = 0.7d = 15 mm.

7. Welded Joints

Welding gives a permanent joint by fusing the metal at the edges. Fillet welds are made on the corners of the lap and T joints; butt welds are made between plates in the same plane.



Weld symbol = arrow + reference line + weld sign

Types and proportions

Joint	Weld	Features
Lap joint	fillet (single or double)	overlap = 5 t (min.), triangular section; size s = 0.7 t; throat = 0.707 s
T joint	fillet (on one or both sides)	the stem plate on the base plate; size s = t
Butt joint	square edge (t < 5 mm)	gap 1 - 2 mm
Butt joint	single V (60° - 90°), double V (X), U, J	root gap 2 mm, root face 2 mm, depth to suit t

Welding symbol (IS 813)

- The symbol is made of the reference line (horizontal), the arrow line (pointing at the joint), the tail (for the process / specification), and the elementary symbol (triangle for fillet, V for butt, etc.).
- Weld on the arrow side: the elementary symbol below the reference line; other side: above it. Weld all round: a circle at the arrow junction; site (field) weld: a flag. The leg size s is at the left of the symbol, and the length at the right, e.g. 6 > 50 = 6 mm fillet, 50 mm long.
- In the conventional representation the weld is shown by the thick line with the hatch / blackened triangle for the section.

8. Keys

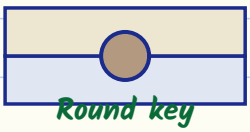
A key is a machine part placed in the keyway of a shaft and hub to prevent relative rotation and to transmit torque (pulleys, gears, couplings).



Taper key
w = d/4, t = 2w/3, taper 1:100



Sunk taper (gib head)
head height = 1.5 t



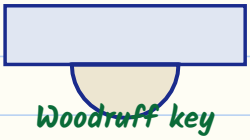
Round key
dia = d/4, half in each



Saddle key
hollow base, no keyway



Feather key
two screws on shaft



Woodruff key
semi-circular, deep seat

Key	Proportions (shaft dia d)	Feature
Sunk (rectangular) key	width w = d/4, thickness t = 2w/3 = d/6, length L = 1.5 d to 2 d	half in the shaft, half in the hub
Taper key	as above, taper 1 in 100 on the top	hub keyway tapered; driven by the hammer
Gib-head taper key	same, with the head 1.5 t high, projected 1.5 t	easy to withdraw
Round key	dia = d/4	pin in the hole drilled half in the shaft, half in the hub (light duty)
Saddle key	w = d/4, t = 2w/3	no keyway in the shaft, the hollow underside grips (friction) - light loads
Feather key	w = d/4, t = 2w/3, fixed by 2 screws on the shaft	the hub slides along the shaft (axial movement)
Woodruff key	semi-circular disc, dia about 2.5 d (from the IS table), thickness b = d/4	deep seat, self adjusting, light duty, automobile and machine tools

Sunk key in the shaft - the end view

- Draw the shaft circle (d) and the hub circle (2d). The keyway is a rectangle w wide in the shaft, depth t/2; the same in the hub. The top of the key is seen with the clearance from

the hub keyway (the taper key fits tightly, but the sunk key has the clearance on top).

- In the sectional front view the key is not hatched, because it is a solid part cut along the length (shown by a plain rectangle).

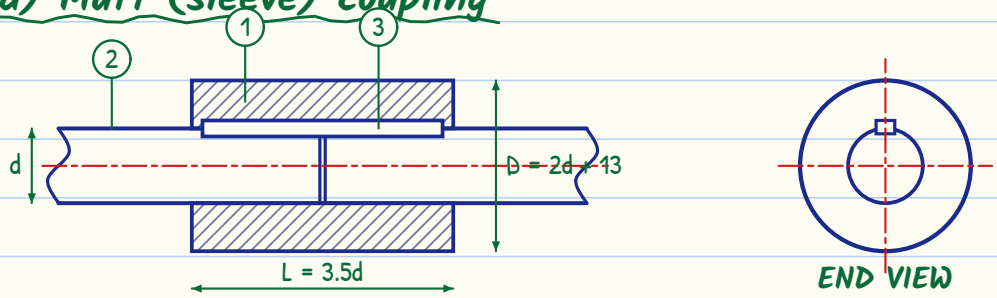
Example

Shaft 40 mm dia: sunk key w = 10 mm, t = 6.7 ~ 7 mm (IS key 12 x 8 for the d = 38 - 44), L = 1.5d = 60 - 80 mm; the keyway depth in the shaft t/2 = 3.5 mm.

9. Couplings - Rigid

A coupling connects two shafts end to end to transmit power. Rigid couplings are used for perfectly aligned shafts.

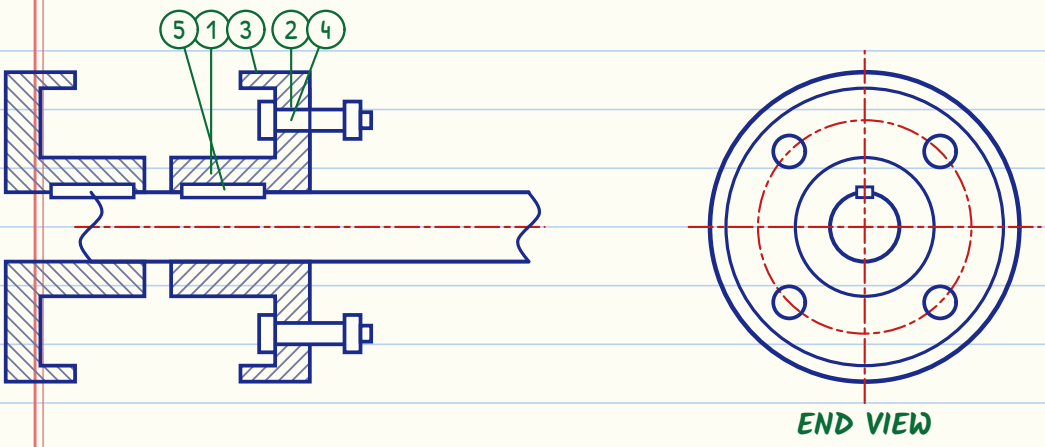
(a) Muff (sleeve) coupling



1 Sleeve (C.I.) 2 Shaft 3 Sunk key (w = d/4)

- A cast iron hollow cylinder (the muff) with a sunk key (one key for the both shafts, same keyway length) over the abutting ends of the two shafts.
- Proportions: outer dia D = 2d + 13 mm; length L = 3.5 d; key w = d/4, t = 2w/3, key length = L/2 on each shaft.
- Draw the front view in the half section: the shafts not hatched, the sleeve and the key shown. End view: the circles d and D, with the key.
- Used for the light duty, small power, the lengths not to be moved. The split muff coupling has two halves joined by bolts.

(b) Flange coupling (protected type)



1 Hub 2 Flange 3 Protecting rim 4 Bolt and nut 5 Key

- Two cast iron flanges with hubs keyed to the shafts, joined by the bolts and nuts through the flange. A spigot (recess) on one flange ensures the alignment. The protecting rim covers the bolt heads and nuts to prevent accidents (the protected type; the unprotected type has no rim).

Part	Proportion (d = shaft dia)
Outer dia of hub	D = 2 d
Length of hub	L = 1.5 d
Flange thickness t	0.5 d
Thickness of the protecting rim	0.25 d
Pitch circle dia of bolts D_1	3 d
Outer dia of the flange D_2	4 d (D_1 + 1 d)
Number of bolts n	3 (d up to 40), 4 (40 - 100), 6 (100 - 180)
Bolt dia d_1	0.5 d / √n (nearest standard)
Key	width d/4, thickness d/6 (2w/3), length = hub length L

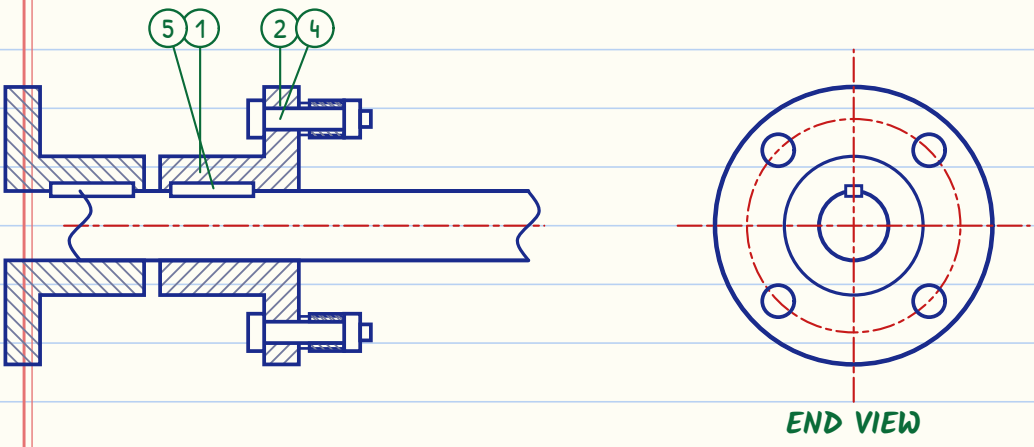
Example

Flange coupling for shaft d = 40 mm: D = 80, L = 60, t = 20, rim 10, D_1 = 120, D_2 = 160, n = 4 bolts of 12 mm dia (M12), key 10 x 7 x 60 (rounded to the standard 12 x 8 x 60).

10. Couplings - Flexible

Flexible couplings allow a small misalignment (angular, parallel) and absorb the shocks and vibrations.

(a) Bushed pin-type flange coupling

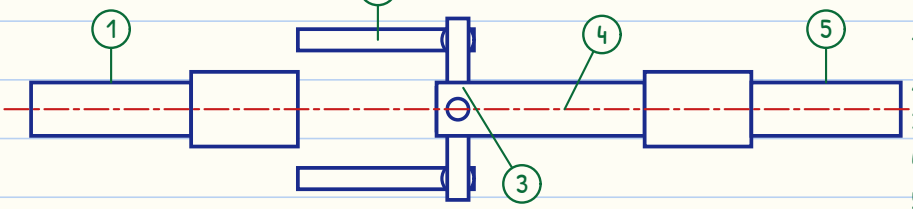


1 Hub 2 Flange 4 Pin with rubber bush 5 Key (no rim)

- Similar to the flange coupling, but the bolts are replaced by the pins with the rubber (leather) bushes; the pins are fixed in one flange and pass with the bush into the other flange. No protecting rim.
- Proportions: as flange coupling; pin dia $d_1 = 0.5\ d\ /\sqrt{n}$, the rubber bush outer dia = d_1 + 2 x (thickness of the bush ~ 6 mm); the number of pins 4 - 6; the gap between the two flanges 5 mm - 8 mm.
- Used for motor - pump / compressor shafts with small misalignment.

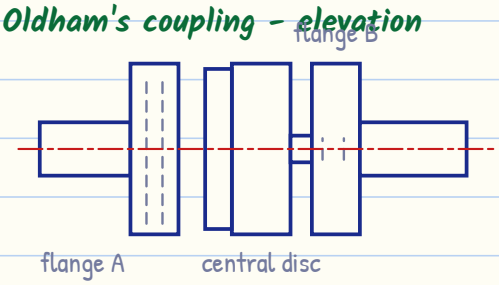
(b) Universal (Hooke's) coupling

Universal (Hooke's) coupling - plan



1 Shaft A
2 Fork A (yoke)
3 Spider (cross)
4 Fork B (turned 90°)
5 Shaft B

Oldham's coupling - elevation



Central disc has a tongue on each face, the two tongues at 90°. Each flange has a diametral slot. Used for shafts with lateral misalignment.

- Connects the shafts with the intersecting axes (angle up to 30° - 40°): two forks (yokes) each keyed to a shaft, joined by a cross (spider) with the four arms (pins), the pins of the spider rotate in the holes of the forks (with the pin and collar).
- Proportions: shaft dia d; fork thickness 0.5 d to 0.6 d, the outer dia of the hub 2d, length of hub 1.5d - 1.75 d, the central block (spider) of the square section 1.2 d - 1.5 d; the pin dia = 0.5 d - 0.75 d, pin length 1.4 d. The fork ends are drawn as the eyes (ears) around the pins; show the front view in half section and the side view at 90°.
- Used in the automobile propeller shaft, the machine tools (milling attachment). The driven shaft's speed is not uniform for one revolution unless the double joint is used.

(c) Oldham's coupling

- For parallel shafts, whose axes are slightly displaced (offset). Two flanges with the diametral slot, one on each shaft, and the central disc with the two tongues at 90° on opposite faces; the tongues slide in the slots. The centre of the disc describes a circle.
- Proportions: flange dia = 3d, thickness 0.5 d - d; slot width = tongue width + 1 mm clearance (about d/2), disc dia 3d, thickness 1.2 d; the hubs 2d dia x 1.5 d long.
- Used in the low speed drive (the central disc has sliding friction, needs lubrication; the centrifugal force at the high speed is high).

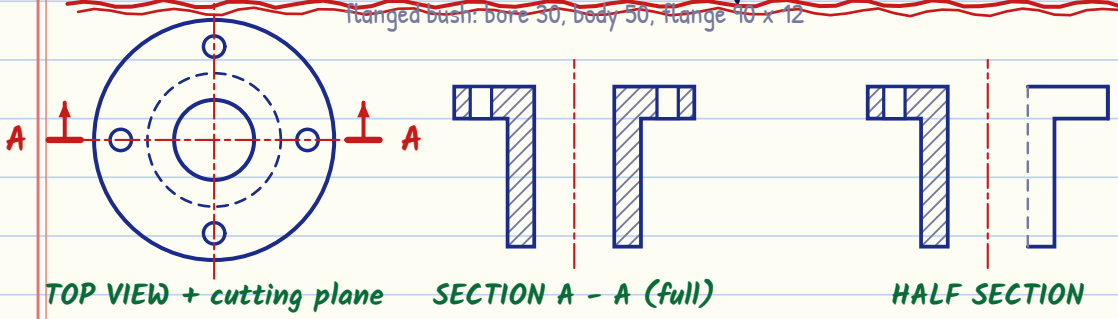
PART B - 3D PACKAGE

11. Solid Modelling - the Workflow

- Sketch: choose the plane (front, top, right), draw the 2D profile with LINE, CIRCLE, ARC; apply the geometric constraints (horizontal, tangent, concentric) and dimensions, so the sketch is fully defined.
- Features: Extrude (boss/base) the profile by the length; Revolve about the axis (shafts, pulleys, bushes); Sweep (springs, threads) along the path; Loft between the sections; Cut (extruded cut, hole wizard); Fillet / chamfer; Pattern (linear, circular) and mirror.
- Assembly: insert the parts, fix the base part, apply the mates (coincident, concentric, parallel, distance, tangent), check the interference and the motion. Drawing (2D from 3D): create the views (front, top, side, isometric), section view, dimensions and the bill of materials (BOM) with balloons.

12. Sectional Views of Complex Machine Parts

Flanged bush: bore 30, body 50, flange 70 x 12



- Create the solid model (e.g. the flanged bush or housing), define the cutting plane (front plane through the axis), use the command Section view / Section cut: the section A - A is generated with the hatching; ribs, webs, spokes, bolts, shafts and keys cut longitudinally are not hatched.
- Types: full section (cutting plane through the entire object), half section (symmetrical parts; one quarter removed: half is seen as the outside and half as the section), offset section (the plane is shifted to pass through the features not in line), revolved / removed section (for arms, ribs, spokes), broken-out (local) section (small part).

- Procedure: (1) model the part, (2) place the cutting plane, (3) view the section - check that the hidden lines are removed, (4) dimension in the drawing mode.

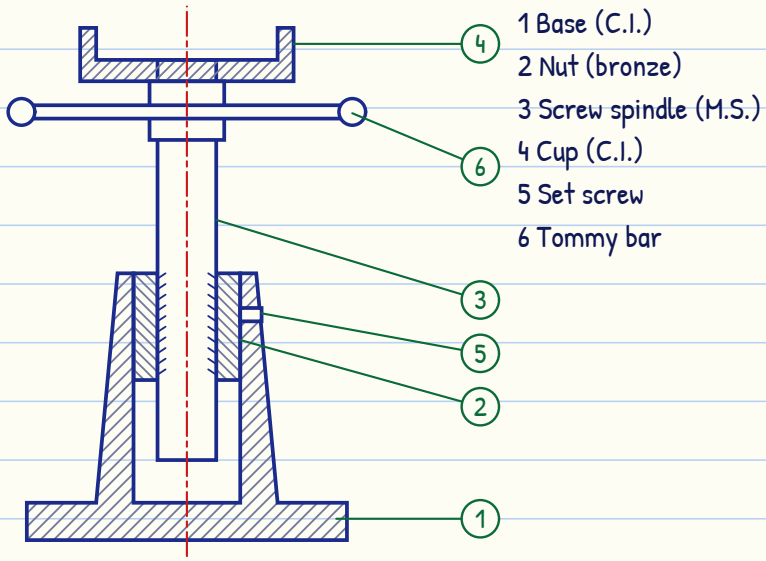
Example

Flanged bush: flange dia 90, thickness 12, bush OD 50 x 40 long, bore 30 H7, 4 holes of Ø10 on PCD 70. Steps: revolve the half profile through 360°; circular pattern 4 holes with the cut; chamfer 1 x 45° on the bore; the front view in the full section A - A, the top view shows the holes and the cutting line.

13. Assembly Drawings (any four)

An assembly drawing shows the parts in the working position, with the part numbers (balloons), the overall dimensions and a BOM (list of parts: item, name, material, quantity). The models of each part are made first, assembled with mates and the 2D views produced (the front view in the section).

(i) Screw jack

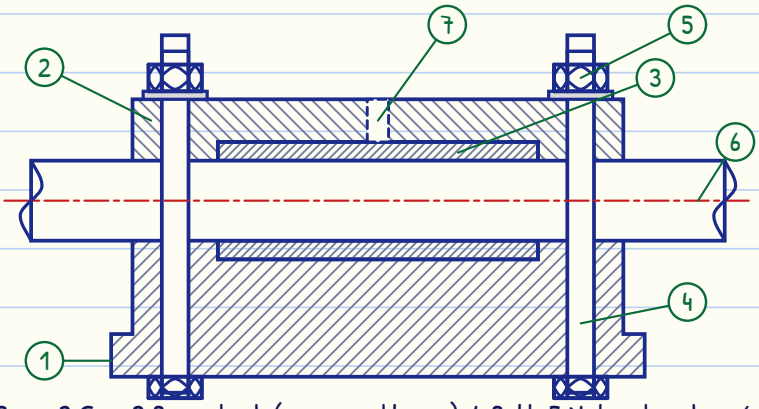


1 Base (C.I.)
2 Nut (bronze)
3 Screw spindle (M.S.)
4 Cup (C.I.)
5 Set screw
6 Tommy bar

- Parts: 1 base (C.I.), 2 nut (bronze) press-fitted in the base, 3 screw spindle (M.S., square / buttress thread), 4 cup (C.I.), 5 set screw (holds the cup but allows it to rotate), 6 tommy bar (handle; M.S.).
- Working: rotating the tommy bar rotates the screw in the fixed nut, so the screw rises and lifts the load on the cup, which does not rotate (the set screw only keeps it from slipping off).

- Typical proportions for the screw dia d: height of the body 2.5 d - 3 d; nut height 1.5 d - 2 d; base dia 2.75 d (dia at the bottom 3 d), thickness of the base 0.3 d; cup dia 2 d; tommy bar dia 0.6 d - d, length 12 d; thread: square, pitch p = d/5 or 0.2 d.

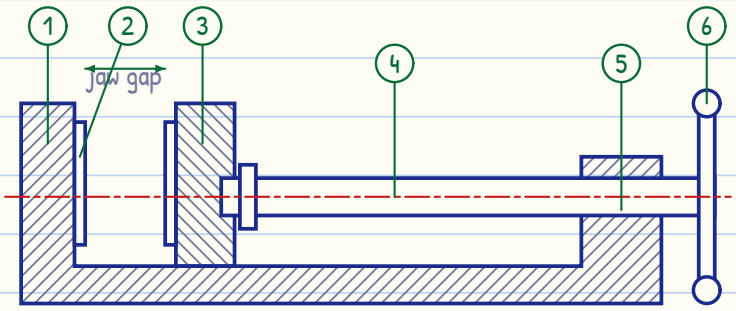
(ii) Plummer block (pedestal bearing)



1 Base 2 Cap 3 Brass bush (upper and lower) 4 Bolt 5 Nut and washer 6 Shaft 7 Oil hole

- Parts: 1 body (C.I.) with the base and bolts holes, 2 cap (C.I.), 3 brass bush (two halves, with the flange or collars to hold it), 4 bolts with nuts (two or four, fix the cap to the body), 5 shaft (supported), 6 oil hole with the cap.
- Proportions for the shaft dia d: length of the bearing = 2 d (1.5 - 2 d), outer diameter of the bush d + 2 x 4 mm; bolt dia 0.2 d to 0.3 d; base length 3.5 d - 4 d, width 2 d, thickness 0.5 d; cap thickness 0.5 d.
- Function: supports the rotating shafts (line shafts), allows the wear of the bush to be taken up; the top half (cap) is removed for the maintenance of the bush.

(iii) Machine vice



1 Body (fixed jaw, C.I.) 2 Jaw plates (steel) 3 Sliding jaw 4 Screw 5 Nut (end block) 6 Handle

- Parts: 1 body (fixed jaw, C.I.), 2 jaw plates (steel, hardened, serrated; countersunk screws), 3 sliding jaw, 4 screw (square thread; with the collar), 5 nut (bronze) in the body

and 6 handle (tommy bar with ball ends).

- Working: the handle rotates the screw, and the sliding jaw moves along the guide in the body; the job is clamped between the two jaw plates. Square thread of the pitch 0.2d (more power, no radial force on the nut).

(iv) Other assemblies in the syllabus

Assembly	Main parts	Modelling points
Lathe tool post	post, rocker, ring, bolt, tool clamp, screw	the rocker on the ring gives the height adjustment; clamping screw and T-slot
Shaping machine tool head	the swivel base, vertical slide, tool post, apron, clapper box and the tool, screw	the clapper block pivots to lift the tool on the return stroke
Tail stock	body, barrel (quill), centre (Morse taper), handwheel, screw, clamp	the quill moves in the body by the screw; the Morse taper hole for the centre
Gate valve	body, bonnet, gland, stem (screw), wedge disc, handwheel, packing	the stem with the thread rises the disc to open; gland and the stuffing box for the sealing
Carburetor	float chamber, float, needle valve, jet, venturi, choke and throttle	the revolve for the venturi; the fuel level by the float
Piston and connecting rod	piston with rings, gudgeon pin, small end, big end with the bolts and bearing shells	assemble with the concentric mates of the pin; circlip groove
Eccentric	sheave, strap (two halves), rod, bolts	the eccentric offset e = half the stroke (throw)
Geneva cam	driver with pin, driven wheel with slots (4, 6, 8)	motion mate (cam); the intermittent rotation
Axle bearing, pipe vice, clamping device	body, bush, cap, screws, jaws, pins	the standard parts from the library (the toolbox) - bolts, nuts, washers

PART C - PRODUCTION DRAWING

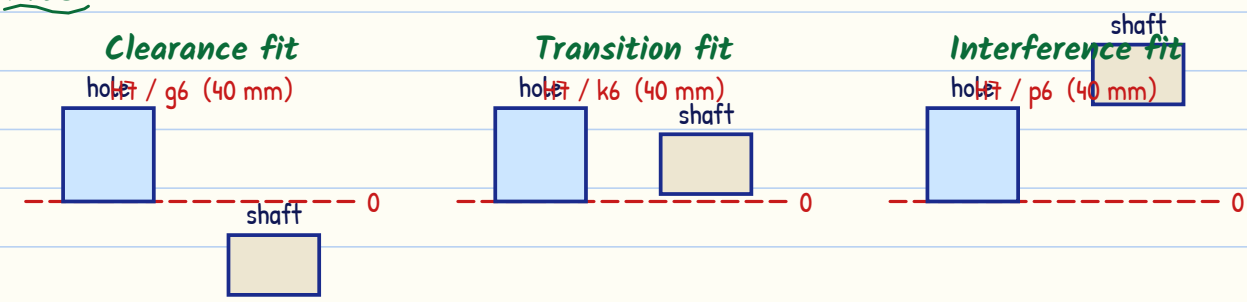
14. Limits, Fits and Tolerances

A part cannot be made to the exact size: the permissible variation is the tolerance. Interchangeability of parts needs the proper selection of limits and fits.

Terms

- Basic (nominal) size: the theoretical size from which the limits are given. Limits: the maximum and minimum permitted sizes. Tolerance = upper limit - lower limit.
- Deviation: the algebraic difference between the size and the basic size. Upper deviation (ES for hole, es for shaft) and lower deviation (EI, ei). Fundamental deviation: the one closest to the zero line, given by the letter (A to ZC hole, a to zc shaft).
- Tolerance grade (IT01 to IT18): the number. Hole basis system: the hole (H, EI = 0) is the constant and the shaft varies; used more as holes are made by the standard drills and reamers.
- Allowance: the intentional difference between the dimensions of mating parts (min. clearance or max. interference).

Fits



max 0.050 mm min 0.009 mm +0.023 mm -0.018 mm max 0.042 mm min 0.001 mm

zero line = basic size; hole basis system (H hole); IT7 = 25 µm, IT6 = 16 µm

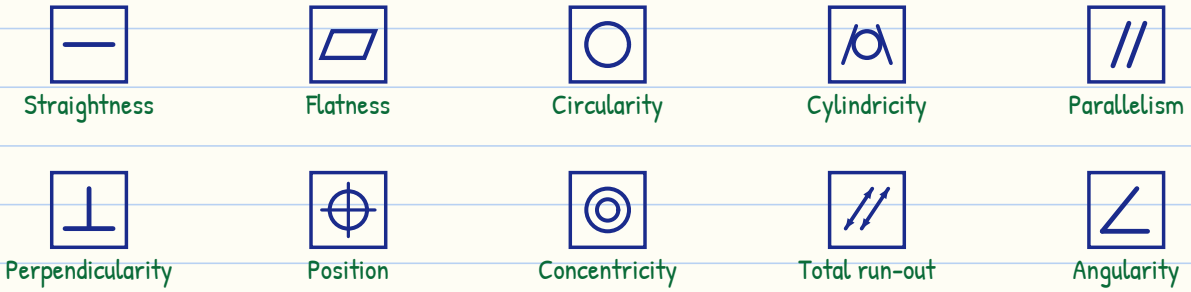
Fit	Condition	Example (hole basis)	Use
Clearance	the shaft is always smaller than the hole	H7 / g6, H8 / f7	sliding fit, bearings, spindles
Transition	clearance or interference, depends on the actual sizes	H7 / k6, H7 / n6	location of pulleys, gears on shaft
Interference (force)	the shaft is always larger than the hole	H7 / p6, H7 / s6	press fit of bushes, bearing races

Example - Ø40 hole basis (IT7 = 25 µm, IT6 = 16 µm)

Fit	Hole	Shaft	Result
H7 / g6	40.000 - 40.025	39.975 - 39.991	max clearance 0.050, min clearance 0.009
H7 / k6	40.000 - 40.025	40.002 - 40.018	max clearance 0.023, max interference 0.018
H7 / s6	40.000 - 40.025	40.043 - 40.059	max interference 0.059, min interference 0.018

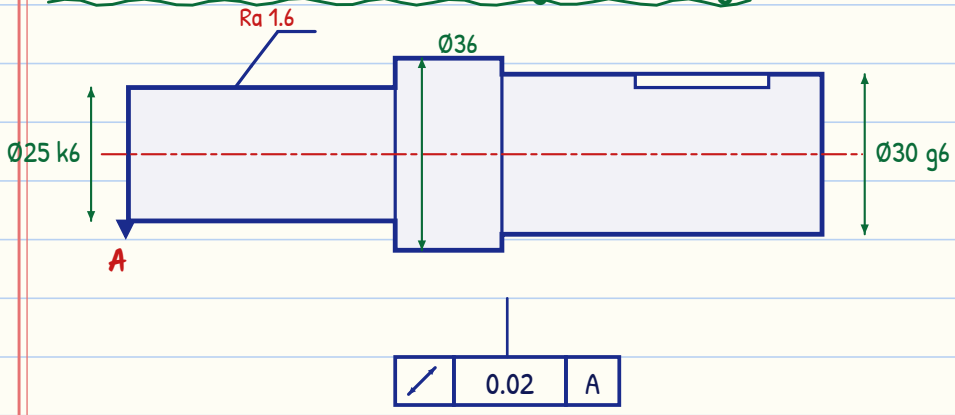
- Tolerance of the hole = 40.025 - 40.000 = 0.025 mm; of the shaft (g6) = 39.991 - 39.975 = 0.016 mm.

Geometric tolerances (GD and T)



- They control the form, orientation, location and run-out; written in the feature control frame: [symbol | tolerance value | datum]. Form: straightness, flatness, circularity, cylindricity (no datum); orientation: parallelism, perpendicularity, angularity; location: position, concentricity, symmetry; run-out: circular and total.
- Surface finish: Ra (μm) values in the symbol: Ra 0.8 grinding / lapped, Ra 1.6 fine turning, Ra 3.2 machining, Ra 6.3 - 12.5 rough machining.

Production (manufacturing) drawing



circular run-out 0.02 with respect to datum A

- Take any four parts of the assembly (e.g. the screw spindle, the nut, the base and the cup of the jack). For each part draw the views necessary to manufacture the part (usually the sectional front view and the end view), with all dimensions (size, location, tolerance), surface finish symbols, geometric tolerances, the material, the title block (name, scale, projection symbol, units) and the general notes (e.g. fillets R3, chamfer 1 x 45°).
- The fits of the mating parts: bush in the body H7 / p6 (press); shaft in the bush H7 / g6 (running); the nut in the base H7 / n6 or H7 / s6 (to prevent rotation). Dimension in the chain, parallel (baseline) or coordinate form; use the datum surface for the machining.

- The Ra of the bearing surface 0.8 – 1.6 µm; the machined faces 3.2 µm; non-machined (casting) surfaces 'rough / as cast'.

Viva - Short Answers

- Why is the hole basis used more? Hole is made by the fixed size tools (drills, reamers) - the shaft is easily turned or ground to any size.
- Difference between stud and bolt? Stud is threaded at both ends, one end permanent; the bolt has the head and is removed completely.
- What is the lead of a double start thread? Lead = 2 x pitch.
- Why is a double strap butt joint stronger? The rivets are in double shear and the load is axial (no bending).
- Which key allows the axial sliding? Feather key. Which key needs no keyway in the shaft? Saddle key.
- Which coupling joins the intersecting shafts? Universal (Hooke's); parallel offset shafts: Oldham's.
- What is an assembly drawing? A drawing with the parts at their working positions, with the item numbers and the parts list; does not give the full dimensions.
- What are the 3D modelling mates? Coincident, concentric, parallel, perpendicular, tangent, distance and angle.